



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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UNIT TEST - 3, NOVEMBER - 2024

Time Allowed : 1.30 Hours]

PHYSICS

[Max. Marks : 35

PART - I

Choose the correct Answer.

5x1=5

Light transmitted by Nicol Prism is

- a) Partially Polarised b) Unpolarised
c) Plane Polarised d) Elliptically Polarised

Two Coherent monochromatic light beams of Intensities I and $4I$ are superposed. The maximum and minimum Possible intensities in the resulting beam are

- a) $5I$ and I b) $5I$ and $3I$ c) $9I$ and I d) $9I$ and $3I$

First diffraction minimum due to a single slit of width 1.0×10^{-5} cm is at 30° . Then wavelength of light used is.

- a) $400 \text{ \AA}$ b) $500 \text{ \AA}$ c) $600 \text{ \AA}$ d) $700 \text{ \AA}$

Emission of electrons by the absorption of heat energy is called ----- emission.

- a) Photoelectric b) field c) thermionic d) secondary

The Threshold wave length for a metal surface whose Photoelectric work function is 3.313 eV is

- a) $4125 \text{ \AA}$ b) $3750 \text{ \AA}$ c) $6000 \text{ \AA}$ d) $2062.5 \text{ \AA}$

PART - II

Answer any Three questions and Question number 10 is compulsory.

3x2=6

What is Photoelectric effect?

Distinguish between interference and diffraction.

Define stopping Potential.

Mention any Four uses of Polaroids.



10. Calculate the distance upto which ray optics is a good approximation for light of wavelength 500 nm falls on an aperture of width 0.5mm.

PART-III

III. Answer any three questions and question No. 15 is compulsory.

11. Write note on continuous X-ray spectra.
12. Differentiate between Fresnel and Fraunhofer diffraction.
13. State laws of photoelectric effect.
14. State and Prove Brewster's law.
15. Calculate the energies of the photons associated with the following radiation (i) Violet light 413 nm and (ii) X-rays of 0.1nm.

PART - IV

IV. Answer the following.

16. a) Explain the Principle and Working of Electron Microscope.

(OR)

- b) (i) Mention any two Characteristic of Photon.
- (ii) Obtain Einstein's photo electric equation with necessary explanation.

17. a) Obtain the equation for bandwidth in Young's double slit experiment.

(OR)

- b) Discuss the experiment to determine the wavelength of monochromatic light using diffraction grating?

18. a) Explain the construction and working of photo emissive cell with neat diagram. Give two uses of it.

(OR)

- b) Prove laws of refraction using Huygen's Principle